
PAVAN TEJA PENUGONDA

SENIOR ENGINEER

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PROFESSIONAL SUMMARY

Experienced **Senior Software Engineer** with 7+ years of expertise in designing and delivering **scalable, high-performance software solutions** across diverse architectures. Authorized to work in the U.S. under OPT status. Proven ability to **translate complex business requirements into robust systems**, leveraging strong skills in **coding, unit testing, and compliance-driven development**. Adept at working across **data, application, and business layers**, ensuring seamless integration and alignment with stakeholder objectives. Hands-on experience in **CI/CD pipelines, story refinement, requirement analysis, and effort estimation** throughout the delivery lifecycle. Skilled in building and maintaining **comprehensive test suites** (integration, regression, performance), analyzing results, and performing **root-cause analysis** to resolve issues efficiently. Demonstrated success in driving **innovation through proof-of-concepts**, applying **machine learning, large language models (LLMs)**, and **retrieval-augmented generation (RAG)** architectures to solve complex challenges. Passionate about **continuous improvement, automation**, and delivering solutions that enhance system reliability and business value.

WEBSITES, PORTFOLIOS, PROFILES

- github.com/Pavantejapenugonda
- linkedin/in/pavantejapenugonda

SKILLS

Technical & Programming Skills AI, ML & Data
Science Compliance & Financial Crime
Methodologies & Delivery

Frameworks & Tools: FastAPI, React, Flask,
PowerApps, Docker, GitHub, Azure DevOps

Cloud Platforms: AWS, Azure, GCP

Languages: Python, Java, C#, JavaScript, SQL,
HTML, CSS

Databases: MSSQL Server, PostgreSQL, MongoDB,
DynamoDB, Chroma, Elasticsearch

DevOps & CI/CD: Pipeline design, build, deploy,
maintain, automated testing

Machine Learning: Random Forest, XGBoost, Deep Learning, Image & Text Classification

Data Processing: ETL optimization, Data validation, Data cleansing & normalization

AML / BSA Compliance, OFAC Sanctions, PEP Screening, EDD

Model Validation, Stress Testing, Gap Assessment

Agile & Scrum, Requirements Gathering, Story Refinement

Continuous Improvement, Root Cause Analysis, Operational Reporting

Generative AI / LLMs: GPT-4, RAG, Fine-tuning, Agentic Workflows

Visualization & Analytics: Dashboards, Predictive Insights, PowerBI

Transaction Monitoring (TM), Case Management (CMT)

Compliance Reporting & Audit Readiness, Regulatory Coverage Analysis

Stakeholder Management, Global Delivery, Backlog Prioritization

WORK HISTORY

Senior Analyst, 02/2025 - Current

EY US LLC – San Antonio, United States

- **Performed AML gap assessments**, designing technical workflows to detect, analyze, and document cross-system discrepancies, improving compliance coverage.
- **Enhanced sanctions monitoring** by optimizing ETL pipelines and developing validation cases aligned with OFAC sanctions lists, strengthening screening accuracy.
- **Developed data validation tools** leveraging Python, SQL, and automation frameworks; collaborated with cross-functional teams to identify bottlenecks and deliver proof-of-concept solutions using **AI and LLMs**.
- **Implemented machine learning models** (Random Forest, XGBoost) for TM alert classification, achieving 91% accuracy and reducing false positives significantly.
- **Created visualization dashboards** to interpret complex AI model outputs, enabling stakeholders to make informed decisions based on predictive insights.
- **Configured and supported CI CMT workflows** across AML workstreams (TM, EDD, PEP/Sanctions), standardizing alert and case management processes end-to-end.
- **Engineered automated scripts** for stress testing and model validation, generating comprehensive reports for OFAC, Red Flags, and Actimize models.
- **Optimized system performance** by identifying workflow bottlenecks and implementing targeted enhancements, reducing processing time and improving SLA adherence.
- Designed an **agentic compliance intelligence system** to assess regulatory coverage gaps across multiple agencies (FDA, EMA, SEC, ISO) using **Defy-orchestrated LLM agents**.

- Implemented a **RAG-based knowledge layer** using regulatory documents, guidance notes, and historical audits, enabling **evidence-grounded GPT-4o reasoning**.
- Developed **coverage scoring and gap analysis agents** in Python to map organizational controls against regulatory clauses.
- Automated compliance reports with **explainable citations**, improving audit readiness and reducing manual review effort by **60–70%**.
- Designed an **agentic KYC documentation system** leveraging **GPT-4 Vision** for identity document understanding, fraud detection, and data extraction.
- Implemented **Python-based documentation clarification and decision-making agents** to resolve ambiguities, detect inconsistencies, and automate approval workflows.
- Built a **FastAPI backend** with asynchronous agent orchestration and a **React + HTML frontend** for real-time reviewer feedback.
- Enabled **explainable KYC decisions** with confidence scores and audit-ready reasoning traces.

Software Engineering Intern, 06/2024 - 12/2024

SS8 Networks – Milpitas, California

- Developed proof-of-concepts (POCs) for media data analysis leveraging large language models (LLMs), including audio (translation, transcription, summarization), image (summarization), and video (summarization).
- Built a service for text analytics, enabling sentiment analysis, word grouping, and hypothesis generation.
- Created multiple microservices to improve scalability.
- Conducted research on OSINT data and developed a Retrieval-Augmented Generation (RAG) model for enhanced search functionality.

Teaching Assistant, 01/2024 - 05/2024

University of Bridgeport – Bridgeport, CT

- Helping professor to prepare assignments, coding documents, starter scripts.
- Helping students to clear the doubts other than regular class sessions.
- Helping professor to grade the assignments submitted by the students.

Senior Software Engineer, 05/2021 - 08/2023

EY GDS – Bangalore, Karnataka

- Leading the Performance Improvement team of a python flask-based KYC project built on microservice architecture.
- Used multiple performance optimization techniques like multi-threading, code, SQL query optimization, Indexing, also AKS configuration changes and improved performance of application by 200%.
- Built an image classification and text detection algorithm for the KYC application.

- Worked in creating automated data cleansing and normalization pipelines using PowerApps, python webapps.
- Implemented logging mechanism across the entire application for better troubleshooting using basic logging module and Distributed Tracing system.

Software Engineer, 02/2020 - 05/2021

Smart Auto Systems – Hyderabad, Telangana

- Improved the performance of the ML algorithm to 87% by using parameter tuning, data cleaning techniques.
- Design and development of single page applications using Flask, SQL SERVER, VueJS, CSS, HTML5.
- Research and developed the image processing solution using existing form recognizer APIs from different cloud services.

Python Developer, 08/2017 - 01/2020

Spry Technologies – Bangalore, Karnataka

- Worked on the data analysis and report generation using PowerBI Dashboards.
- Build Web automation scripts to test the application functionality using Selenium.
- Followed best practices with code development, maintenance and testing to improve solution design.

EDUCATION

Master's: AI, 08/2023 - 12/2024

University of Bridgeport - Bridgeport, CT

- 3.9 GPA

ATHLETICS INVOLVEMENT

- Treasurer, Student Entrepreneur Club (SEC), 2023, Present
- Member, Mathematics, Biology Club, 2024, Present

PUBLICATIONS

- Microstructure, hardness and wear behavior of AZ31 Mg alloy - fly ash composites produced by friction stir processing, ScienceDirect, 27
- Surface Engineering of ZE 41 Mg Alloy by Friction Stir Processing: Effect of Process Parameters on Microstructure and Hardness Evolution, ScienceDirect, 8

COMMUNITY SERVICE & VOLUNTEER WORK

- Volunteered multiple on-campus events like Home Coming event (Pillow Stuffing Stall), Student Involvement Event (Helped ISC club), Fundraising Event.
- Volunteered events conducted by Orphanages for raising funds, Food distribution, teaching children.